

MAL3040 — Statistics
Quiz 1 — with answers

Department of Mathematics, Indian Institute of Technology Jodhpur

B.Sc.–B.Ed. III year

Time: 50 minutes

Maximum marks: 15 (+2 bonus)

The answers are printed in **this colour** below each part, briefly and with most of the working left out. If your own answer disagrees with one of them, go back to your script and find where the two of you parted company.

Formulas. For n observations $x_1, \dots, x_n$:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2, \quad \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2, \quad s = \sqrt{s^2}.$$

Question 1. Consider the six numbers

8, 5, 9, 11, 8, 7.

(a) Find $\sum x_i$ and $\sum x_i^2$. **[2]**

Ans. $\sum x_i = 48$, $\sum x_i^2 = 404$.

(b) Find the sample mean. **[1]**

Ans. $\bar{x} = 48/6 = 8$.

(c) Arrange the numbers in increasing order and write down the median. **[1]**

Ans. 5, 7, 8, 8, 9, 11; median = $(8 + 8)/2 = 8$.

(d) Find the sample standard deviation. **[1]**

Ans. $\sum (x_i - \bar{x})^2 = 404 - 6(8)^2 = 20$, $s^2 = 20/5 = 4$, $s = 2$.

[5 marks]

Question 2. Twenty students wrote a test marked out of 50. Their marks are shown in the stem and leaf plot below.

1		2	2	5			
2		1	3	4	6	7	9
3		0	0	2	4	5	6 8
4		1	3	5	7		

Key: the row 1 | 2 stands for a mark of 12.

(a) Make a frequency table showing the number of students in each of the four groups 10–19, 20–29, 30–39 and 40–49. **[1]**

Ans. 3, 6, 7, 4 students, adding to 20.

(b) Draw a bar chart of these four frequencies. Label both axes. **[1]**

Ans. Four bars of heights 3, 6, 7, 4, with the groups on the horizontal axis and the number of students on the vertical axis.

(c) Write down the median mark, and say in one line how you found it from the plot. **[1]**

Ans. Median = 30. With $n = 20$ it is the average of the tenth and eleventh marks; counting leaves gives $3 + 6 = 9$ before the third row, so both fall on that row and are 30 and 30.

(d) Find the sum of the twenty marks, and hence the mean mark. **[1]**

Ans. Row totals 39, 150, 235, 176 give $\sum x_i = 600$; mean = $600/20 = 30$.

[4 marks]

Question 3. A small shop has five workers. Their monthly salaries, in thousands of rupees, are

10, 12, 14, 16, 18.

- (a) Find the mean salary and the median salary of the five workers. [1]

Ans. Mean = $70/5 = 14$; median = 14 (both in thousands of rupees).

- (b) The owner also draws a salary from the shop. It is 50 thousand rupees. Add the owner to the list, so that there are now six salaries. Find the new mean and the new median. [1]

Ans. Mean = $120/6 = 20$; median = $(14 + 16)/2 = 15$.

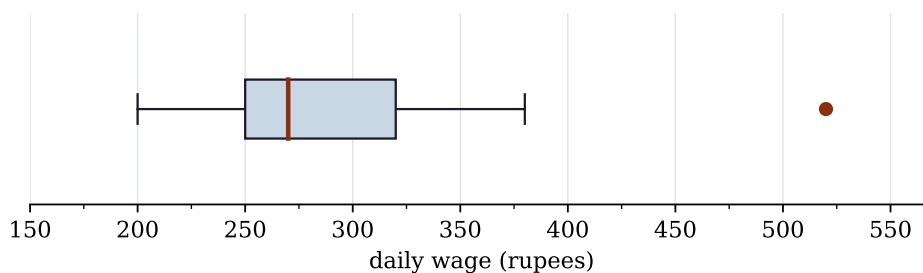
- (c) A newspaper wants to print one number describing the “typical” salary in this shop. Should it print the mean or the median? Give a reason in one or two sentences. [1]

Ans. The median. The owner’s salary lifted the mean by 6 but the median by only 1, and the mean of 20 is above five of the six salaries, so it describes nobody in the shop.

[3 marks]

Question 4. The box plot below shows the daily wages of twenty workers.

The whiskers reach the extreme wages lying within $1.5 \times (Q_3 - Q_1)$ of the box; anything beyond is drawn as a separate dot.



- (a) Read off the median, the first quartile Q_1 and the third quartile Q_3 . [1]

Ans. $Q_1 = 250$, median = 270, $Q_3 = 320$ rupees.

- (b) Is the data roughly symmetric, or is it stretched out towards the higher wages? Give one reason for your answer, based on what you can see in the picture. [1]

Ans. Stretched out towards the higher wages. The median line sits nearer the left edge of the box than the right, and the right whisker is the longer of the two.

- (c) One wage is drawn as a separate dot, away from the rest. What does this tell you about that worker? Does the dot mean that somebody made a mistake while writing down the wages? [1]

Ans. That worker earns far more than the others, about 520 against a next highest of about 380. It does not mean a mistake was made; the wage may well be genuine.

[3 marks]

Bonus question. Optional. These marks are over and above the fifteen marks of the paper.

Both parts below were asked and answered in the first class.

- (a) We asked two questions about the heights of the students in this class.

Question A: A student is to be chosen at random from this class. What is the probability that the chosen student is over five feet tall?

Question B: Mr. X is one particular student in this class. What is the probability that Mr. X is over five feet tall?

One of these two questions makes sense and the other does not. Say which is which, and justify your answer in one or two sentences. [1]

Ans. Question A makes sense; Question B does not. In A the student has not yet been chosen, so the outcome is still open. Mr. X’s height is already a fact, and measuring him would settle it; not knowing it does not make it random.

- (b) A card is about to be drawn at random from a well shuffled pack of 52 cards. Write down the probability that the card drawn will be a spade.

The card is now drawn, and the student who drew it has looked at it but has shown it to nobody. Does the number you wrote still describe the card in her hand? Justify your answer in one or two sentences. [1]

Ans. $13/52 = 1/4$. No. That number describes the draw before it happens. Once the card is drawn it either is a spade or it is not; our not knowing which does not make it random, and the answer is certainly not $1/2$.